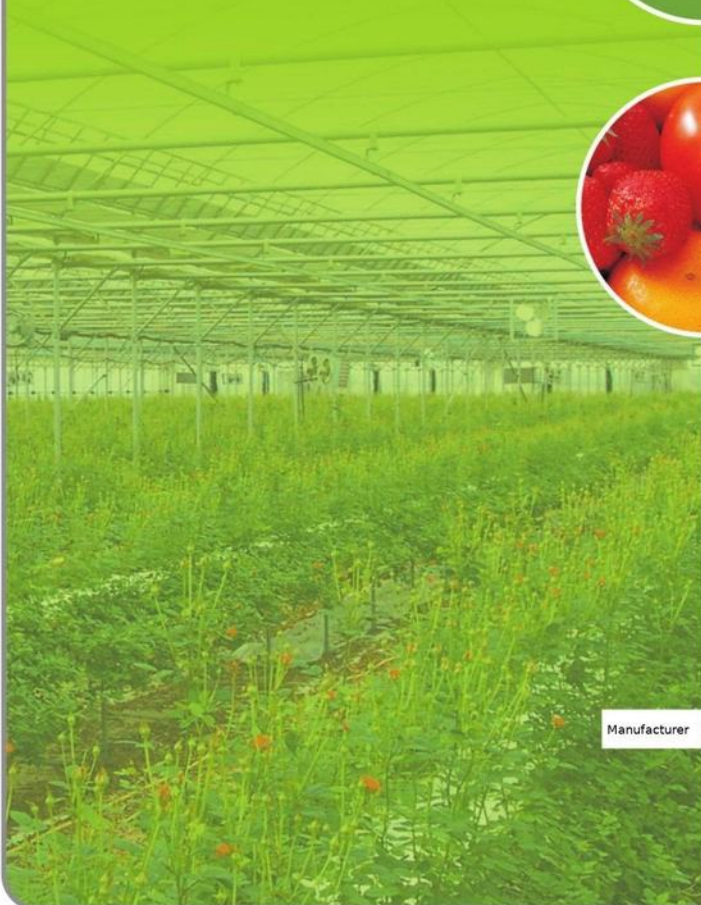


Leader in fertigation and nutrient solution systems

Hydroponic Agrow



Manufacturer : **Hangaram Agricultural Development (Oct. 2024)**

Gangwon-do, Wonju-si, Hoje-myeon, Umugae-ro 274

TEL : (033) 747-5114, 731-7377

FAX : (033) 746-1909, 731-9877

28 years of nutrient feeder manufacturing technology!

The leader in fertigation, nutrient solutions and feeders — Agrow!

Hydroponic **Agrow**

Introducing our best products!

· Nutrient Feeder — Agrow-Perfect (Full option/Basic)

Agrow-7000 PLUS (Full option/Basic)

Agrow-6000 PLUS (Full option/Basic)

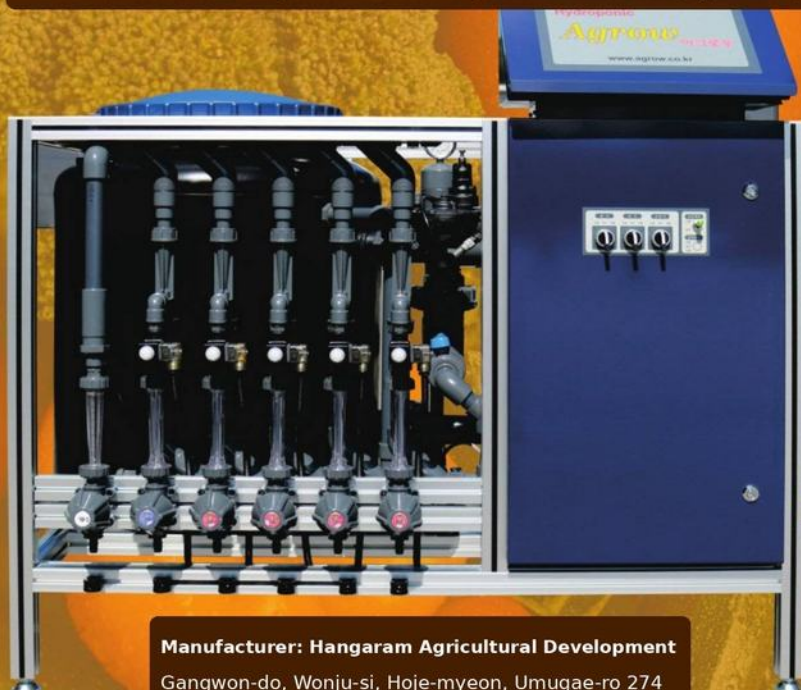
Agrow-7000 (Full option/Basic)

Agrow-6000 (Full option/Basic)

Agrow-5000 Master (Full option/Basic)

Agrow-5000 Magic (Full option/Basic)

· EC-controlled fertigation unit — Agrow-2000 Star (Type A/B)



Manufacturer: Hangaram Agricultural Development

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www.agrow.co.kr

※ The appearance and structure of products in this catalog may be partially changed without prior notice for process improvement.

Complete P.I.D. nutrient control with proportional valves Direct nutrient supply without a mixing tank



2023 New

Nutrient Feeder

Agrow - Perfect <Basic/Option>

Features

- Complete P.I.D. nutrient control using proportional valves
- Direct nutrient supply without a mixing tank — minimal control deviation
- Nutrient = A/B/C/D/E, acid/base, aux. fertilizer = up to 8 channels (option)
 - Standard = A/B, acid/base and auxiliary fertilizer = 4 channels
- 6 nutrient control programs run independently according to set operating periods
- Of the 6 nutrient control programs:
 - Group EC/pH control and 3-step accumulated radiation control = 3 programs
 - Valve EC/pH control and 3-step accumulated radiation control = 3 programs
 - Group EC/pH control and 3-step moisture content control = 1 program
- Radiation coefficient can be applied to accumulated radiation programs
- Valve auxiliary fertilizer = controlled by flow (1 L = option) and time (min:sec)
- EC-1 (control)/EC-2 (monitor) + pH-1 (control)/pH-2 (monitor) sensors — standard
 - Option = EC-3 (reprocess)/EC-4 (drain)/EC-5 (substrate)
 - pH-3 (reprocess)/pH-4 (drain)/pH-5 (substrate)
- Operation by external signal = linked with integrated environment control
- Standalone PC connection and mobile (web) control = communication (option)
- Irrigation pump selectable = 5.5 / 7.5 / 10 HP
- 10.2" touchscreen LCD
- Safety = 3-phase 380V — phase loss / low level / irrigation & raw-water pump fault
 - EC / pH / flow alarm
- Other functions are the same as Agrow-7000plus

Model No. HPA-9001



Best functions and content

Agrow-7000plus 종합현황-1									
작동 방법	대기	전역시간	전역	전역	전역	전역	전역	전역	전역
작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램
작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램
작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램
작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램
작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램
작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램
작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램
작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램	작동 프로그램



10.2" touchscreen



Mobile remote



PC program

Nutrient Feeder

Agrow - 7000 Plus <Basic/Option>

Features

- PC & smartphone: linked operation, remote control = 10 ch.
- 5 independent programs (radiation-only — 3 / moisture & external sensors — 2)
- One radiation sensor drives 3 independent programs by accumulated radiation
- In the 3 radiation programs, radiation [-Ec:00%] and moisture [+00%] coefficients apply
- Nutrient — 6 (A~F) + acid/base + auxiliary fertilizer = 8 / irrigation zones = 16
- Nutrient supply mode per program: proportional-integral / combined
- In combined mode, per-component (A~F) offset supply is possible (10%~90%)
- Select nutrients per 5 programs: 6 (A~F) / acid-base / auxiliary fertilizer
- Irrigation volume per 5 programs = time (min:sec) / flow (0000 L)
- Auxiliary fertilizer unit = selectable from valves 1~16
- Substrate moisture control & temperature display = dedicated programs (pro-4/pro-5)
- Weight control & external environment control = dedicated programs (pro-4/pro-5)
- 5 programs run independently by sensors, time interval and set time
- After irrigation in each of the 5 programs, valves can be flushed with raw water
- For multi-crop growing: mixing tank drain and nutrient preparation on program change
- Per-valve delay time can be entered when irrigation zones change (00 sec)
- Standard EC sensors (2): EC-1 = control / EC-2 = monitors EC-1
- Standard pH sensors (2): pH-1 = control / pH-2 = monitors pH-1
- In-program EC/pH: alarm limits adjustable; safety device activates at limits
- Sensor value calibration for EC-1/2/3, pH-1/2/3 and radiation sensor
- Data: daily per 5 programs — irrigation count / per 16 valves — flow and irrigation time
- Option: irrigation flow meter = switches to automatic control on meter failure
- Option: drain flow meter = totals and records drain; avg EC/pH of 16 valves stored for 1 year
- Option: EC-3 / pH-3 = reprocessing sensors; reprocessing range can be set, alarm and shut-off when reprocessed mixture outside the range enters
- Option: EC-4 / pH-4 = drain sensors; EC-5 / pH-5 = substrate sensors; radiation sensor

Model No. HPA-7000 PLUS



HP Agrow

Leader in fertigation, nutrient solutions and supply · Agrow

Product for mixed crops



Nutrient Feeder

Agrow - 6000 Plus <Option>

Features

- PC & smartphone: linked operation, remote control = 10 ch.
- 4 independent programs (radiation — 3 / substrate moisture & external sensors — 1)
- One radiation sensor drives 3 independent programs by accumulated radiation
- In the 3 radiation programs, radiation [-Ec:00%] and moisture [+00%] coefficients apply
- Nutrient = 4 (A~D) + acid/base + auxiliary fertilizer = 6 / irrigation zones = 16
- Per 4 programs: select from 4 nutrients (A~D) / acid-base / auxiliary fertilizer
- Per 4 programs: irrigation volume = time (min:sec) / flow (0000 L)
- Auxiliary fertilizer unit = selectable from valves 1~16
- Substrate moisture control & temperature display = dedicated program (pro-4)
- Weight control & external environment control = dedicated program (pro-4)
- 4 programs run independently by sensors, time interval and set time
- After irrigation in each of the 4 programs, valves can be flushed with raw water
- Standard: 1 EC sensor / 1 pH sensor
- In-program EC/pH: alarm limits adjustable; safety device activates at limits
- Sensor value calibration for EC-1/2, pH-1/2 and radiation sensor
- Data: daily per 4 programs — irrigation count / per 16 valves — flow and irrigation time
- Average EC/pH values of 16 valves stored for 1 year and viewable
- Option: irrigation flow meter = switches to automatic control if the flow meter fails
- Option: drain flow meter = totals and records drain; radiation sensor
- Option: EC-2 / pH-2 = reprocessing sensors; reprocessing range can be set, alarm and shut-off when reprocessed mixture outside the range enters
- Option: EC-3 / pH-3 = drain sensors

Model No. HPA-6000 PLUS

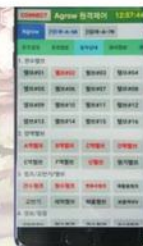


Farmers' choice — No.1 in sales

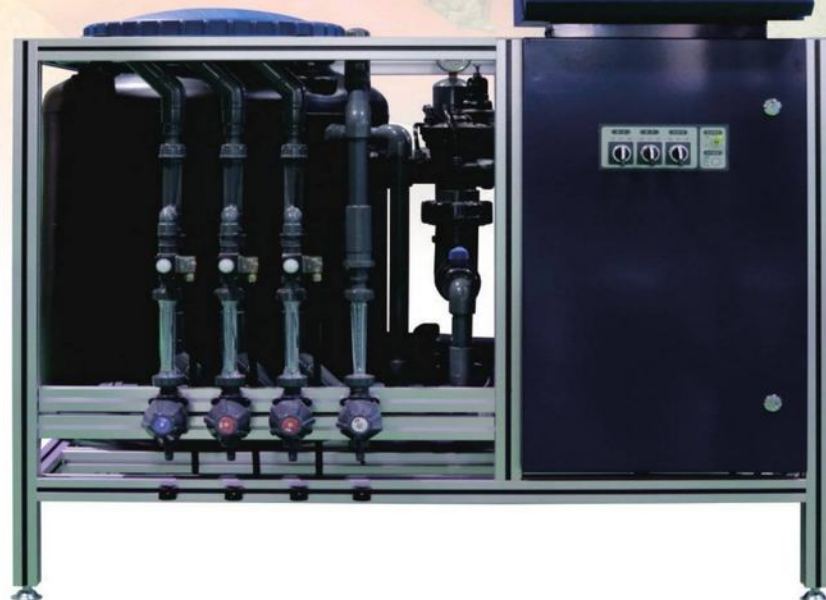
Agrow-6000plus 통합현황-1									
작물 방법	수동	남용시간	00:00	현재 재량사량	0	W/m	설정EC	0.0	mS
작물 프로그램	프로그램 1	작물별비	비율계량	현재 EC	0.0	mS	설정pH	0.0	mg/L
비율계량 (100%)	0.0	비율계량	0.0	현재 pH	13.9	mg/L	비율계량	0.0	mg/L
비율계량 (100%)	0.0	비율계량	0.0	비율계량	0.0	mg/L	비율계량	0.0	mg/L
비율계량 (100%)	0.0	비율계량	0.0	비율계량	0.0	mg/L	비율계량	0.0	mg/L
비율계량 (100%)	0.0	비율계량	0.0	비율계량	0.0	mg/L	비율계량	0.0	mg/L
비율계량 (100%)	0.0	비율계량	0.0	비율계량	0.0	mg/L	비율계량	0.0	mg/L
비율계량 (100%)	0.0	비율계량	0.0	비율계량	0.0	mg/L	비율계량	0.0	mg/L
비율계량 (100%)	0.0	비율계량	0.0	비율계량	0.0	mg/L	비율계량	0.0	mg/L



10.2" touchscreen



Mobile remote



PC program

Nutrient Feeder

Agrow - 6000 Plus <Basic>

Features

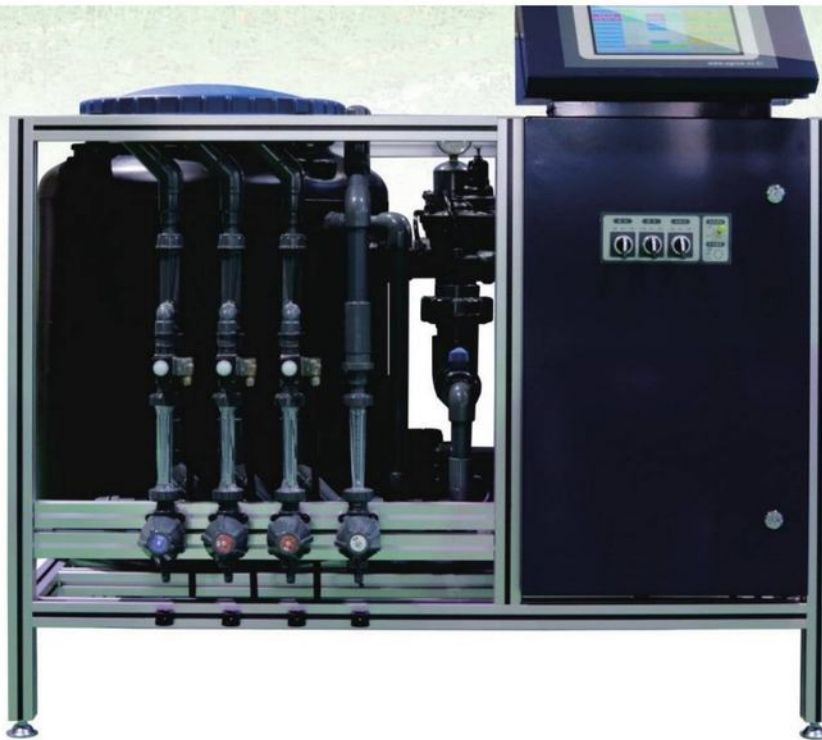
- PC & smartphone: linked operation, remote control = 10 ch.
- 4 independent programs (radiation — 3 / substrate moisture & external sensors — 1)
- One radiation sensor drives 3 independent programs by accumulated radiation
- In the 3 radiation programs, radiation [-Ec:00%] and moisture [+00%] coefficients apply
- Nutrient = 4 (A~D) + acid/base + auxiliary fertilizer = 6 / irrigation zones = 16
- Per 4 programs: select from 4 nutrients (A~D) / acid-base / auxiliary fertilizer
- Per 4 programs: irrigation volume = time (min:sec) / flow (0000 L)
- Auxiliary fertilizer unit = selectable from valves 1~16
- Substrate moisture control & temperature display = dedicated program (pro-4)
- Weight control & external environment control = dedicated program (pro-4)
- 4 programs run independently by sensors, time interval and set time
- After irrigation in each of the 4 programs, valves can be flushed with raw water
- Standard: 1 EC sensor / 1 pH sensor
- In-program EC/pH: alarm limits adjustable; safety device activates at limits
- Sensor value calibration for EC-1/2, pH-1/2 and radiation sensor
- Data: daily per 4 programs — irrigation count / per 16 valves — flow and irrigation time
- Average EC/pH values of 16 valves stored for 1 year and viewable
- Option: irrigation flow meter = switches to automatic control if the flow meter fails
- Option: drain flow meter = totals and records drain; radiation sensor
- Option: EC-2 / pH-2 = reprocessing sensors; reprocessing range can be set, alarm and shut-off when reprocessed mixture outside the range enters
- Option: EC-3 / pH-3 = drain sensors

Model No. HPA-6000 PLUS



**With 20 years of nutrient feeder export know-how,
packed with convenient functions and content**

Agrow-7000 종합현황-1									
농약동작비교-3월	P-1	P-2	P-3	P-4	P-5	현재	0	W/m	
작동 방법	대기	작업시간				설정EC		mS	
농. 프로그램	수소-1대	작동범위				현재	2.4	mS	
EC-2(pH-2)	수소	원재계수				설정pH		mS	
EC-3(pH-3)	수소	설정계				현재	7.3	mS	
농. 공급시간	0000:00	분:초				설정pH		mS	
1일 농. 주:량	0L	1일관수량	0.00			비상			
관수	대역					정지			
Agrow-7000-2									
Agrow-7000-2									



Nutrient Feeder

Agrow - 7000 <Basic/Option>

Features

- 5 independent programs (radiation — 3 / moisture & ext. sensors — 2)
- One radiation sensor drives 3 independent programs by accumulated radiation
- In the 3 radiation programs, radiation [-Ec:00%] and moisture [+00%] coefficients apply
- Nutrient — 6 (A~F) + acid/base + auxiliary fertilizer = 8 / irrigation zones = 16
- Nutrient supply mode per program: proportional-integral / combined proportional
- In combined mode, per-component (A~F) offset supply is possible (10%~90%)
- Select nutrients per 5 programs: 6 (A~F) / acid-base / auxiliary fertilizer
- Irrigation volume per 5 programs = time (min:sec) / flow (0000 L)
- Option: flow meter = auto switch to other control if the meter fails in flow-based supply
- Moisture & external sensor programs = program assignment per sensor (P-4/P-5)
- 5 programs run independently by external sensors, time interval and set time
- After irrigation in each of the 5 programs, valves can be flushed with raw water
- For multi-crop growing: mixing tank drain and nutrient preparation on program change
- Per-valve delay time can be entered when irrigation zones change (00 sec)
- Standard EC sensors (2): EC-1 = control / EC-2 = monitor
- Standard pH sensors (2): pH-1 = control / pH-2 = monitor
- Option: EC-3 / pH-3 = reprocessing sensors; reprocessing range can be set, alarm and shut-off when reprocessed mixture outside the range enters
- In-program EC/pH: alarm limits adjustable; safety device activates at limits
- Sensor value calibration for EC-1/2/3, pH-1/2/3 and radiation sensor
- Data: daily per 5 programs — irrigation count / per 16 valves — flow and irrigation time
16 valves — average EC/pH stored 1 year and viewable. Radiation sensor

Model No. HPA-7000





Nutrient Feeder

Agrow - 6000 <Basic/Option>

Features

- 4 independent programs (radiation — 3 / moisture & ext. sensor — 1)
- One radiation sensor drives 3 independent programs by accumulated radiation
- In the 3 radiation programs, radiation [-Ec:00%] and moisture [+00%] coefficients apply
- Nutrient = 4 (A~D) + acid/base + auxiliary fertilizer = 6 / irrigation zones = 16
- Per 4 programs: select from 4 nutrients (A~D) / acid-base / auxiliary fertilizer
- Per 4 programs: irrigation volume = time (min:sec) / flow (0000 L)
- Option: flow meter = auto switch to other control if the meter fails in flow-based supply
- Moisture & external sensor program = program assignment per sensor (P-4)
- 4 programs run independently by ext. sensors, interval and set time (20 times)
- After irrigation in each of the 4 programs, valves can be flushed with raw water
- Standard: 1 EC sensor / 1 pH sensor
- Option: EC-2 / pH-2 = reprocessing sensors; reprocessing range can be set, alarm and shut-off when reprocessed mixture outside the range enters
- In-program EC/pH: alarm limits adjustable; safety device activates at limits
- Sensor value calibration for EC-1/2, pH-1/2 and radiation sensor
- Data: daily per 4 programs — irrigation count / per 16 valves — flow and irrigation time
Average EC/pH values of 16 valves stored for 1 year and viewable

Model No. HPA-6000



New 2024



Agrow-5000Master 종합현황-1									
2024년03월09일 화 10시01분06초									
작동방법	P1	P2	P3	대기	-	-	-	현재일시량	
작동별브	간격	작동 프로그램	1					414 W/m ²	
배액 EC-2	01-통작중	남은시간	00:04					EC 설정/현재(mg)	1.2 / 1.1
배액 pH-2	0.0	배액 pH-2	0.0					pH 설정/현재(logm)	5.8 / 5.9
관수율/온도	0%	배액 유량	0 L						
1일 공급시간	0000:06분초	1일 관수량	0 L						
보관수량	1 회	교반기	통작 ON						
제한경보	해제	-	-					비상	중합현황-2
경보	해제	-	-					정지	

Nutrient Feeder

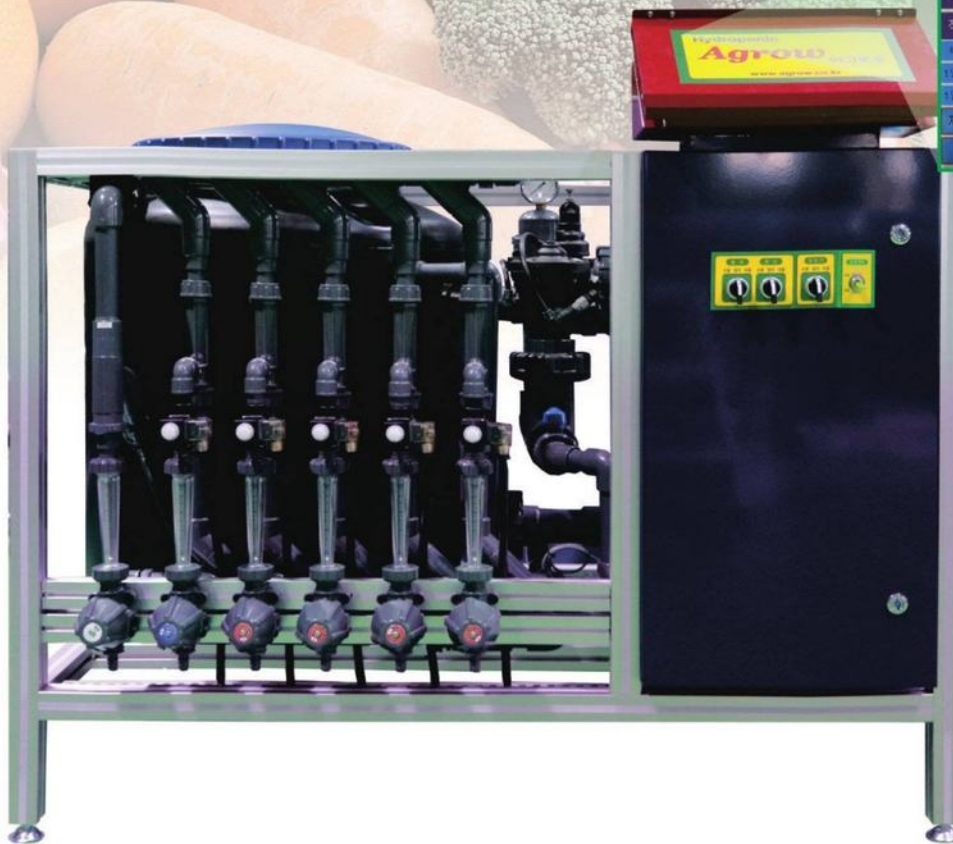
Agrow - 5000 Master <Basic/Option>

Model No. HPA-5002

Features

- Consists of 3 independent nutrient control programs
- Programs: radiation (2-step) — 2 / moisture — 1
- EC-pH group control = 2 / EC-pH valve control = 1 program
- Standard: 7" touchscreen LCD
- Standard nutrient set = A / B, acid/base and auxiliary fertilizer
- Optional nutrient set = A / B / C / D, acid/base and auxiliary fertilizer
- Standard sensors = control EC-1 / control pH-1
- Options = drain EC-2 / drain pH-2 / radiation sensor / flow sensor
moisture sensor / raw-water pump / irrigation solenoid valve — sold separately
- Irrigation zones = 12; supply by radiation / moisture / time interval
supply by set time / operation by external signal possible
- Auxiliary fertilizer by direct injection — no effect on mixing tank EC / pH
- EC-1 / pH-1 sensors interchangeable with Agrow-6000 / 7000
- Irrigation pump = stainless, 3 HP (2.2 kW) horizontal / continuous mixing
- Safety = 3-phase 380V — phase loss / low level / pumps / EC, pH, flow alarms





Agrow-5000Magic 종합현황-1				20011101 10:01 일 오후 01:01분 19.5	
작동방법	P1	P2	P3	작동 프로그램	현재물수량
작동밸브	대기	작동 프로그램	-	-	0 W/m ³
해역 EC-2	0.0 mS	해역 pH-2	0.0 -logH+	EC 설정/현재(mS)	- / 0.0
1일 용량시간	0000:00분초	1일 용수량	0 L	pH 설정/현재(-logH+)	- / 14.4
1일 관수횟수	0 회	교반기	-	비상정지	종합현황-2
제한경보	해제	-	-	-	-
경보	해제	-	-	-	-

Nutrient Feeder

Agrow - 5000 Magic <Basic/Option>

Model No. HPA-5001

Features

- Consists of 3 nutrient control programs
- Accumulated radiation control is possible per program
- Standard: 7" screen — touchscreen LCD
- Nutrient = A/B, acid/base and auxiliary fertilizer = 4 (standard)
Nutrient = A/B/C/D, acid/base and auxiliary fertilizer = 6 (option)
- EC-1 (control) + pH-1 (control) = standard
- Option = EC-2 (drain) + pH-2 (drain)
- 12 irrigation zones per program; volume = flow (flow meter optional) / time (min:sec)
- Auxiliary fertilizer per program — direct injection on/off = standard
- Agrow-3000/5000 and Agrow-6000/7000 EC sensors interchangeable = pH sensor shared
- Irrigation pump = 3 HP stainless + continuous mixing = standard
- Safety = 3-phase 380V — phase loss / low level / pumps / alarms
EC / pH / flow alarm
- Other functions are the same as Agrow-5000



New 2024



EC-Controlled Fertigation Unit Agrow - 2000 Star <Type B>

■ Mixed supply (Continouns Mixing) - Model No. HPA-2005B

Features

- First fertigation unit with 2 built-in independent programs
- Radiation / moisture control — 2 independent operating programs
- Group EC program = 1 / valve EC program = 1
- Irrigation zones = 12 (per valve — time and flow supply)
- Independent EC control for each of the 12 irrigation zones
- Built-in programs for raw water supply before and after irrigation
- Built-in irrigation valve delay program before irrigation
- Irrigation zones = 12; supply by radiation / moisture / time interval
 - Set-time supply / built-in operation by external signal
- Standard: 7" touchscreen LCD
- Standard sensor = EC control sensor
- Options = radiation / moisture / flow sensor / auxiliary fertilizer unit
 - raw-water pump / irrigation solenoid valve — ordered separately
- Irrigation pump = stainless, 2 HP (1.5 kW) horizontal / continuous mixing supply
- Safety = irrigation pump / raw-water pump fault / EC / flow alarm



New 2024



EC-Controlled Fertigation Unit

Agrow- 2000 Star <Type A>

■ **Direct injection** (Direct-Injection) - **Model No. HPA-2005A**

Features

- First fertigation unit with 2 built-in independent programs
- Radiation / moisture control — 2 independent operating programs
- Group EC program = 1 / valve EC program = 1
- Irrigation zones = 12 (per valve — time and flow supply)
- Independent EC control for each of the 12 irrigation zones
- Autonomous proportional A/B dosing by target vs. average EC deviation — economical
- Nutrient = separate switches for A and B for independent A/B management
- Irrigation zones = 12; supply by radiation / moisture / time interval

Set-time supply / built-in operation by external signal

- Standard: 7" touchscreen LCD
- Standard sensor = EC control sensor
- Options = radiation / moisture / flow sensor / auxiliary unit / irrigation solenoid valve — separate
- Irrigation pump = stainless, 2 HP (1.5 kW) horizontal / direct injection supply
- Safety = irrigation pump fault / EC / flow alarm

